



E3.S PCIe Gen5 SSD

Version 1.4
Feb. 04, 2026

T7P5

1.92TB, 3.84TB, 7.68TB, 15.36TB

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Revision History

Revision	Date	Description	Editor
1.0	May. 05, 2025	Initial release	
1.1	Nov. 04, 2025	Modify Ordering Information	
1.2	Dec. 02, 2025	Modify Model Number	
1.3	Jan. 12, 2026	Modify Product List Model Name	
1.4	Feb. 04, 2026	Modify Model Number	

Form Factor	OP	Model Number	Capacity	LBA Count (512B Size)
E3.S	OP7	TRUSTA T7P5-001T9T5E3S01D040	1920GB	3,750,748,848
		TRUSTA T7P5-003T8T5E3S01D040	3840GB	7,501,476,528
		TRUSTA T7P5-007T6T5E3S01D040	7680GB	15,002,931,888
		TRUSTA T7P5-015T3T5E3S01D040	15360GB	30,005,863,778
	OP28	TRUSTA T7P5-001T6T7E3S01D040	1600GB	3,125,627,568
		TRUSTA T7P5-003T2T7E3S01D040	3200GB	6,251,233,968
		TRUSTA T7P5-006T4T7E3S01D040	6400GB	12,502,446,768
		TRUSTA T7P5-012T8T7E3S01D040	12800GB	25,004,893,538

Features	Environmental Specifications	
PCI Express Gen5 x 4 Lane (Single Port)	Temperature Range	
PCI Express Base Specification Rev. 5.0 Compliant	- Operating (by SMART)	0°C ~ 70°C
NVM Express Specification Rev. 2.0c Compliant	- Non-Operating	-40°C ~ 85°C
Power Loss Protection (by On-board Circuit)	Shock (Non-Operating, Half Sine)	1500G / 0.5ms
Thermal Throttling	Vibration	
SMBus Basic Management (Out of Band)	- Non-Operating, Sine	20G / 20 ~ 2000Hz
Secure Boot, Crypto Erase	- Non-Operating, Random	3.13Grms / 5 ~ 800Hz
Security AES GCM 256bits, SHA2-512, RSA-4096	- Operating, Random	1.8Grms / 5 ~ 500Hz
TCG Opal SSC v2.01	Humidity (Non-Operating)	5 ~ 95%

Drive Configuration		Reliability Specifications	
Interface	PCIe Gen5x4	UBER	< 1 Error per 10 ¹⁷ Bits Read
NAND	SanDisk BiCS6 eTLC	MTBF	2.5 Mhrs
Protocol	NVMe v2.0c	Endurance	1.0 DWPD / 3.0 DWPD (5 Years Assumed)
Sector Size	512B, 512B+8B, 4KiB, 4KiB+8B, 4KiB+64B		
		Data Retention	3 Months (40°C)

Performance (Typ.)		Quality of Service (Latency)	
Data Transfer Rate (128KB Data Size, Up to)		Average Latency (4KB Data Size, Sustained, QD1, Typical)	
Sequential Read	13,100 MB/s (512B LBAF)	Random Read	75 us
	13,500 MB/s (4KiB LBAF)	Random Write	10 us
Sequential Write	10,400 MB/s (512B LBAF)		
	10,300 MB/s (4KiB LBAF)		
Data I/O Speed (4KB Data Size, Sustained, Up to)			
Random Read	3,100 KIOPS (512B LBAF)		
	3,200 KIOPS (4KiB LBAF)		
Random Write	470 KIOPS (OP7)		
	870 KIOPS (OP28)		

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1.0 Overview

1.1 General Description

The T7P5 is an NVMe 2.0c-compatible SSD featuring a PCIe Gen5 x4 interface in the EDSFF E3.S form factor. It delivers industry-leading performance and consistent Quality of Service (QoS), along with exceptional reliability for PCIe NVMe-based systems. Available in capacities of 1920GB, 3840GB, 7680GB, and 15360GB, the T7P5 is ideal for high-performance, data-intensive applications.

1.2 Product List

Following table has E3.S form factor SSD model numbers with 8 capacities.

Type	Model Number	Capacity
E3.S	TRUSTA T7P5-001T6T7E3S01D040	1600GB
	TRUSTA T7P5-001T9T5E3S01D040	1920GB
	TRUSTA T7P5-003T2T7E3S01D040	3200GB
	TRUSTA T7P5-003T8T5E3S01D040	3840GB
	TRUSTA T7P5-006T4T7E3S01D040	6400GB
	TRUSTA T7P5-007T6T5E3S01D040	7680GB
	TRUSTA T7P5-012T8T7E3S01D040	12800GB
	TRUSTA T7P5-015T3T5E3S01D040	15360GB

Table 1-1 E3.S Products

2.0 Mechanical Specifications

2.1 Physical Dimensions and Weights

Parameter	Unit	1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.36TB
Width	mm	76.0 ± 0.25							
Length	mm	112.75 ± 0.40							
Thickness	mm	7.5 + 0.20/ -0.50							
Weight	g	< 122							

Table 2-1 Physical Dimension and Weight

2.2 Mechanical Drawing

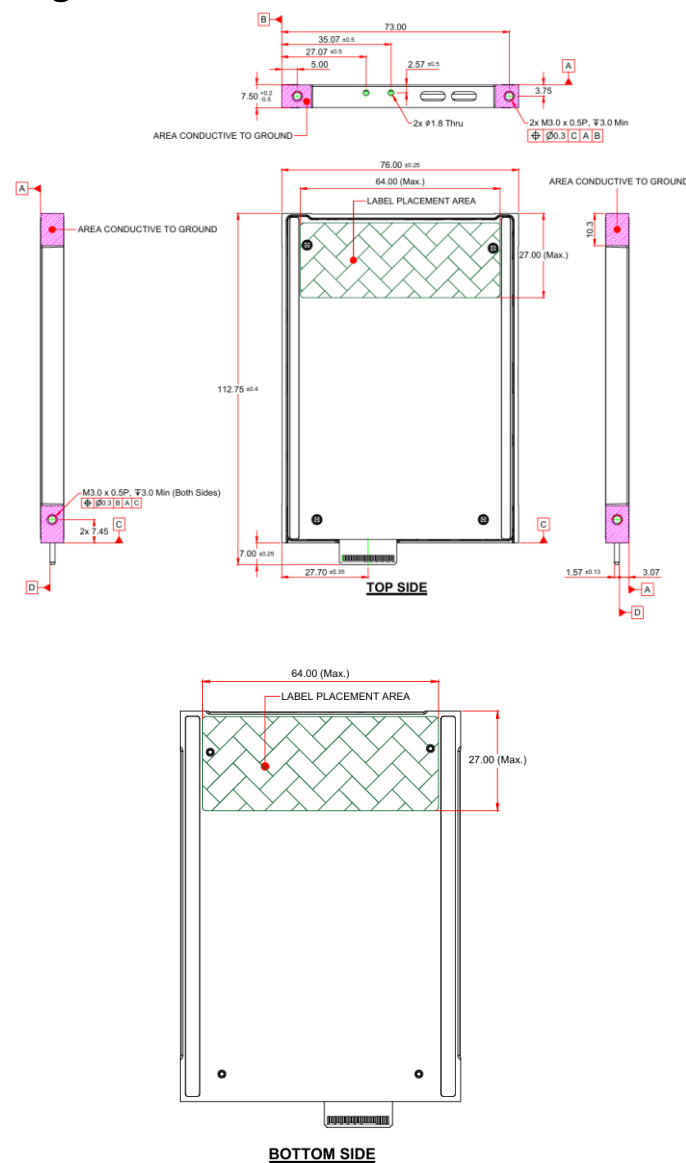


Figure 2-1 Dimension with 1T Case Assembly

3.0 Product Specifications

3.1 Interface Specification

E3.S form factor pin map follows Enterprise and Datacenter SSD Pin and Signal Specification.

Pin#	Assignment	Description	Pin#	Assignment	Description
B1	12V	Primary Power	A1	GND	Ground
B2	12V	Primary Power	A2	GND	Ground
B3	12V	Primary Power	A3	GND	Ground
B4	12V	Primary Power	A4	GND	Ground
B5	12V	Primary Power	A5	GND	Ground
B6	12V	Primary Power	A6	GND	Ground
B7	MFG		A7	SMBCLK	SMBus Clock
B8	RFU		A8	SMBDAT	SMBus Data
B9	DUALPORTEN#		A9	SMBRST#	
B10	PERST0#		A10	LED	
B11	3.3V AUX		A11	PERST1#/CLKREQ#	
B12	PWRDIS	Power Disable	A12	PRSNT0#	
B13	GND	Ground	A13	GND	Ground
B14	REFCLKn0	PCle Reference Clock -	A14	REFCLKn1	PCle Reference Clock -
B15	REFCLKp0	PCle Reference Clock +	A15	REFCLKp1	PCle Reference Clock +
B16	GND	Ground	A16	GND	Ground
B17	PETn0	PCle Transmit – (lane0)	A17	PERn0	PCle Receive – (lane0)
B18	PETp0	PCle Transmit + (lane0)	A18	PERp0	PCle Receive + (lane0)
B19	GND	Ground	A19	GND	Ground
B20	PETn1	PCle Transmit – (lane1)	A20	PERn1	PCle Receive – (lane1)
B21	PETp1	PCle Transmit + (lane1)	A21	PERp1	PCle Receive + (lane1)
B22	GND	Ground	A22	GND	Ground
B23	PETn2	PCle Transmit – (lane2)	A23	PERn2	PCle Receive – (lane2)
B24	PETp2	PCle Transmit + (lane2)	A24	PERp2	PCle Receive + (lane2)
B25	GND	Ground	A25	GND	Ground
B26	PETn3	PCle Transmit – (lane3)	A26	PERn3	PCle Receive – (lane3)
B27	PETp3	PCle Transmit + (lane3)	A27	PERp3	PCle Receive + (lane3)
B28	GND	Ground	A28	GND	Ground

Table 3-1 E3.S Connector Pin Map

3.2 Capacity

OP	Drive Capacity	LBA Count
OP7	1920GB	3,750,748,848
	3840GB	7,501,476,528
	7680GB	15,002,931,888
	15360GB	30,005,863,776
OP28	1600GB	3,125,627,568
	3200GB	6,251,233,968
	6400GB	12,502,446,768
	12800GB	25,004,893,538

Table 3-2 LBA Count for Each Capacity

3.3 Performance

To measure the performance, following test environment was used. These platforms were used to measure sequential/random performance, latency and QoS.

Mainboard	ASUS PRIME Z690-A
CPU	Intel i9-12900K
DRAM	64GB
O/S	Ubuntu 20.04.5 LTS
I/O Engine	Linux native driver and Kyber I/O scheduler
Test Tool	FIO (3.35)

Table 3-3 Workstation Specification

3.3.1 Sequential Performance

Performance	Unit	LBAF	1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.36TB
Sequential Read (128KB)	MB/s	512B	13,100	13,100	13,100	13,100	13,100	13,100	13,100	13,100
		4KiB	13,500	13,500	13,500	13,500	13,500	13,500	13,500	13,500
Sequential Write (128KB)	MB/s	512B	3,300	3,300	6,900	6,900	10,400	10,400	10,400	10,400
		4KiB	3,300	3,300	6,900	6,900	10,300	10,300	10,300	10,300

Table 3-4 Sustained Max. Sequential Throughput

NOTE:

Sequential throughput, Thread Count = 1, Queue Depth = 128, IO Size = 128KB

3.3.2 Random Performance

Performance	Unit	LBAF	1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.68TB
Random Read (4KB)	KIOPS	512B	2,000	2,000	3,100	3,100	3,100	3,100	3,100	3,100
		4KiB	2,000	2,000	3,200	3,200	3,200	3,200	3,200	3,200
Random Write (4KB)	KIOPS	512B & 4KiB	180	180	360	360	360	470	470	470

Table 3-5 Random Performance

3.3.3 IOPS Consistency

Performance	1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.36TB
Random Read (4KB)	> 95%	> 95%	> 95%	> 95%	> 95%	> 95%	> 95%	> 95%
Random Write (4KB)	> 95%	> 95%	> 95%	> 95%	> 95%	> 95%	> 95%	> 95%

Table 3-6 IOPS Consistency

NOTE:

IOPS consistency (%) = (99.9% IOPS) / (Average IOPS) x 100. (Queue Depth = 32)

3.3.4 Latency

Performance	Unit	1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.36TB
Random Read Latency	us	75							
Random Write Latency	us	10							

Table 3-7 Average Latency

NOTE:

1) Thread Count = 1, Queue Depth = 1, IO Size = 4KB

3.3.5 Quality of Service

Performance		Unit	1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.36TB
QoS 99.9%	Random Read (4KB, QD = 1)	us	90	90	90	95	95	90	95	90
	Random Read (4KB, QD = 64)	us	255	255	205	205	180	180	180	180
	Random Write (4KB, QD = 1)	us	15	15	15	15	15	15	15	15
	Random Write (4KB, QD = 64)	us	360	635	250	320	180	310	180	310

Table 3-8 Quality of Service (QoS)

NOTE:

1) Thread Count = 1, Queue Depth = 1 & 64, IO Size = 4KB

3.3.6 Power on Ready Time

Description	Unit	1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.36TB
After Normal Shutdown	sec	6							
After Unsolicited Shutdown	sec	20							

Table 3-9 Power on Ready Time per Shutdown Types

3.4 Electrical Specifications

3.4.1 Absolute Maximum Voltage Ratings

Values shown are stress ratings only. Functional operation outside normal operating values is not implied. Extended exposure to absolute maximum ratings may affect reliability.

Description	Unit	1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.36TB
Max. Voltage range for Vin	V	-0.2 ~ 14.0							

Table 3-10 Absolute Max. Rating

3.4.2 Supply Voltage

Operating voltage is 12V, $\pm 10\%$.

Description	Unit	1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.36TB
Min. Operating Voltage	V	10.8							
Max. Operating Voltage	V	13.2							

Table 3-11 Min/Max Operating Voltage

3.4.3 Inrush Current

Inrush current is important in that SSD module can start up without overcurrent protection situation. To measure the inrush current, following test environment was used.

- Test Equipment:
 - Power module N6753A in DC Power analyzer Keysight 6705C
 - Oscilloscope Tektronix MDO3024, 30A AC/DC Current Probe -TCP0030A
 - Sample Quantity: 3 pcs
- Test procedure:
 - Temperature condition: 25 °C (Room Temperature)
 - Prior to the starting, all sample drives must pass the test which write and read the entire disk sequentially.
 - Check all power rails are in fully discharged state, especially PLP capacitors, and then supply host power source.
 - Measure voltage and current profile of main 12 V power rail and check peak current level.
 - Maximum power on inrush current must be under 1.3A (15.6W) after host power on.

Description	Unit	1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.36TB
Inrush Current	A	< 1.3							

Table 3-12 Inrush Current per Capacity

3.4.4 Power Consumption

Mainboard	ASUS PRIME Z690-A								
CPU	Intel i9-12900K								
DRAM	64GB								
O/S	Ubuntu 20.04.5 LTS								
Kernel	5.4.0-155-generic								
Measurement Equipment	Quarch PPM, QTL1999								
Test Method	Sequential Write/Read, Random Write/Read and IDLE state (using FIO 3.35)								
Active	Maximum Average Power with 1s window								
Idle	Average Power with 1s window								
Power Consumption	Unit	1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.36TB
Active Read (Max Average)	W	14		19		21		21	
Active Write (Max Average)	W	12		18		23		23	
Idle (Typ.)	W	5		5		5		5	

Table 3-13 Power Consumption

3.5 Environmental Conditions

Temperature is T_C, Composite Temperature.

Environmental Item		1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.36TB
Temperature	Operating	0°C ~ 70°C							
	Non-Operating	-40°C ~ 85°C							
Humidity	Non-Operating	5% ~ 95%							
Shock (Half Sine)	Non-Operating	1500 G / 0.5ms							
Vibration (Sine)	Non-Operating	20 G / 20 ~ 2000Hz							
Vibration (Random)	Non-Operating	3.13 Grms / 5 ~ 800Hz							
	Operating	1.8 Grms / 5 ~ 500Hz							

Table 3-14 Environmental Specification

3.6 Reliability

Reliability Item	Unit	1.6TB	1.92TB	3.2TB	3.84TB	6.4TB	7.68TB	12.8TB	15.36TB
MTBF	Hours	2.5Mhrs							
UBER	-	1 sector per 10^{17} bits read							
Data Retention	-	3 months @ 40°C (EOL)							

Table 3-15 Reliability

Reliability Item	1.92TB	3.84TB	7.68TB	15.36TB	1.6TB	3.2TB	6.4TB	12.BT
DWPD	1.0				3.0			

Table 3-16 DWPD

4.0 Support Command Sets

4.1 Admin Command Set

Opcode	Command Name
00h	Delete I/O Submission Queue
01h	Create I/O Submission Queue
02h	Get Log Page
04h	Delete I/O Completion Queue
05h	Create I/O Completion Queue
06h	Identify
08h	Abort
09h	Set Feature
0Ah	Get Feature
0Ch	Asynchronous Event Request
0Dh	Namespace Management
10h	Firmware Commit
11h	Firmware Image Download
15h	Namespace Attachment
80h	Format NVM
81h	Security Send
82h	Security Receive
84h	Sanitize
C0h ~ FFh	Vendor Specific

Table 4-1 Admin Commands Lis

4.1.1 Identify Device Command

Identify Controller Data Structure

Bytes	Field	Description	Value
1:0	VID	PCI Vendor ID	1CC1
3:2	SSVID	PCI Subsystem Vendor ID	1CC1
23:4	SN	Serial Number (ASCII)	Variables
63:24	MN	Model Number (ASCII)	Variables ex) TRUSTA T7P5-007T6T5E3S01D040
71:64	FR	Firmware Revision	Variables
72	RAB	Recommended Arbitration Burst	2h
75:73	IEEE	IEEE OUI	707C18
76	CMIC	Controller Multi-Path I/O and Namespace Sharing Capabilities Bit 3: 0b - not support Asymmetric Namespace Access Reporting Bit 2: 0b - Controller is associated with a PCI Function. Bit 1: 0b - Device has a single controller Bit 0: 0b - Device has a single NVM subsystem port	0h
77	MDTS	Maximum Data Transfer Size 9h: 2MB	9h
79:78	CNTLID	Controller ID (CNTLID)	1h
83:80	VER	Version	00020000h

87:84	RTD3R	RTD3 Resume Latency	1C9C380h
91:88	RTD3E	RTD3 Entry Latency	4C4B40h
95:92	OAES	Optional Asynchronous Events Supported Bit 9: 1b - supports the Firmware Activation Notices event. Bit 8: 1b - supports the Namespace Attribute Notices event and the associated Changed Namespace List log page.	300h
99:96	CTRATT	Controller Attributes Bit 19: 1b – supports the Flexible Data Placement capability (refer to TP4146 Flexible Data Placement 2022.11.30 Ratified.pdf) Bit 9: 1b – supports reporting of a UUID List Bit 7: 1b - supports reporting of Namespace Granularity	00280h
101:100	RRLS	Read Recovery Levels Supported	0h
111	CNTRLTYPE	Controller Type 1h – I/O Controller	1h
127:112	FGUID	FRU Globally Unique Identifier	"0000000000000000 0000000000000000 000"
129:128	CRDT1	Command Retry Delay Time 1	0h
131:130	CRDT2	Command Retry Delay Time 2	0h
133:132	CRDT3	Command Retry Delay Time 3	0h
257:256	OACS	Optional Admin Command Support Bit 5: 1b – support Directives Bit 3: 1b - support the Namespace Management capability Bit 2: 1b - support the Firmware Commit and Firmware Image Download Bit 1: 1b - support Format NVM Bit 0: 1b – support Security Send and Receive	5Fh
258	ACL	Abort Command Limit (zero based value) (Maximum number of concurrently outstanding Abort commands)	3h
259	AERL	Asynchronous Event Request Limit (zero based value) (Maximum number of concurrently outstanding Asynchronous Event Request commands)	Fh
260	FRMW	Firmware Updates Bit 4: 1h - support firmware activation without a reset Bits 3:1 – Number of firmware slots Bit 0: 0h - Slot 1 is read/write	14h
261	LPA	Log Page Attributes Bit 3: 1h – support the Telemetry Host-Initiated and Telemetry Controller-Initiated log pages and sending Telemetry Log Notices Bit 2: 1h – support extended data for the Get Log Page Command Bit 1: 1h – support the Command Supported and Effects log page	1Eh
262	ELPE	Error Log Page Entries (Number of Error Information log entries stored by controller)	3Fh
263	NPSS	Number of Power States Support (zero based value)	4h
264	AVSCC	Admin Vendor Specific Command Configuration Bit 0: 1h - all Admin Vendor Specific Commands use the format defined in nvme specification	1h
265	APSTA	Autonomous Power State Transition Attributes (APSTA) 0h: not support	0h

267:266	WCTEMP	Warning Composite Temperature Threshold 15Eh: 77°C	15Eh
269:268	CCTEMP	Critical Composite Temperature Threshold 166h: 85°C	166h
271:270	MTFA	Maximum Time for Firmware Activation 0h: undefined	30
275:272	HMPRE	Host Memory Buffer Preferred Size 0h: not support	0h
279:276	HMMIN	Host Memory Buffer Minimum Size 0h: not support	0h
295:280	TNVMCAP	Total NVM Capacity	- 1920GB: 1920383410176 Bytes -- 3840GB: 3840755982336 Bytes - 7680GB: 7681501126656 Bytes - 15360GB: 15363002253312Bytes
311:296	UNVMCAP	Unallocated NVM Capacity	0h
315:312	RPMBS	Replay Protected Memory Block Support 0h: not support	0h
317:316	EDSTT	Extended Device Self-test Time 0h: not support	Ch
318	DSTO	Device Self-test Options 0h: not support	1h
319	FWUG	Firmware Update Granularity 1h: 4KB	1h
321:320	KAS	Keep Alive Support 0h: not support	0h
323:322	HCTMA	Host Controlled Thermal Management Attributes 0h: not support	0h
325:324	MNTMT	Minimum Thermal Management Temperature 0h: not support	0h
327:326	MXTMT	Maximum Thermal Management Temperature 0h: not support	0h
331:328	SANICAP	Bit 31-30: 01b - Media is not additionally modified by the NVMe controller after sanitize operation completes successfully Bit 29: 0b - supports the No-Deallocate After Sanitize bit in a Sanitize command Bit 2: 1b – supports the Overwrite sanitize operation Bit 1: 1b - supports the Block Erase sanitize operation Bit 0: 1b - supports the Crypto Erase sanitize operation	0x400000007h
335:332	HMMINDS	Host Memory Buffer Minimum Descriptor Entry Size	0h
337:336	HMMAXD	Host Memory Maximum Descriptors Entries	0h
339:338	NSETIDMA X	NVM Set Identifier Maximum	0h
341:340	ENDGIDMA X	Endurance Group Identifier Maximum	0h
342	ANATT	ANA Transition Time	0h
343	ANACAP	Asymmetric Namespace Access Capabilities	0h
347:344	ANAGRPM AX	ANA Group Identifier Maximum	0h
351:348	NANAGRPI D	Number of ANA Group Identifiers	0h
355:352	PELS	Persistent Event Log Size 18h: 1.5MB	18h

512	SQES	Submission Queue Entry Size 66h: Max 64B / Min 64B	66h
513	CQES	Completion Queue Entry Size 44h: Max 16B / Min 16B	44h
515:514	MAXCMD	Maximum Outstanding Commands 0h: no limit	0h
519:516	NN	Number of Namespaces	- 1920GB: 20h - 3840GB: 40h - 7680GB: 80h - 15360GB: 80h
521:520	ONCS	Optional NVM Command Support Bit 7: 0h – not support the Verify Bit 6: 1h – support the Timestamp feature Bit 5: 0h – not support Reservations Bit 4: 1h – support the Save field in the Set Feature & the Select field in the Get Feature Bit 3: 1h – support the Write Zeros Bit 2: 1h – support the Dataset Management Bit 1: 1h – support the Write Uncorrectable Bit 0: 0h – not support Compare	5Eh
523:522	FUSES	Fused Operation Support Bit 0: 0h not support	0h
524	FNA	Format NVM Attributes Bit 2: 1h – support cryptographic erase as part of the secure erase Bit 1: 0h – support the secure erase per namespace basis Bit 0: 0h - support the Format per namespace basis	4h
525	VWC	Volatile Write Cache Bit 2:1 - The Flush command supports the NSID field set to FFFFFFFFh Bit 0: 0h - No VWC present	6h
527:526	AWUN	Atomic Write Unit Normal (zero based value)	depending on namespaces (Default=0)
529:528	AWUPF	Atomic Write Unit Power Fail (zero based value)	depending on namespaces (Default=0)
530	ICSVSCC	I/O Command Set Vendor Specific Command Configuration Bit 0: 1h - all NVM Vendor Specific Commands use the format defined in nvme specification	1h
531	NWPC	Namespace Write Protection Capabilities	0h
533:532	ACWU	Atomic Compare & Write Unit 0h – not support	0h
535:534	CDFS	Copy Descriptor Formats Supported Bit 0: 0h does not support Copy Descriptor Format 0h	0h
539:536	SGLS	SGL Support - not support SGL	0h
543:540	MNAN	Maximum Number of Allowed Namespaces	0h
1023:768	SUBNQN	NVM Subsystem NVMe Qualified Name	
529:528	AWUPF	Atomic Write Unit Power Fail (zero based value)	depending on namespaces (Default=0)

Table 4-2 Identify Controller Data Structure

Identify Namespace Data Structure

(Single Namespace with full capacity Example)

Bytes	Field	Description	Value
07:00	NSZE	Namespace Size	1920GB: DF8FE2B0h 3840GB: 1BF1F72B0h 7680GB: 37E3E92B0h 15360GB: 6FC7D25B0h
15:08	NCAP	Namespace Capacity	1920GB: DF8FE2B0h 3840GB: 1BF1F72B0h 7680GB: 37E3E92B0h 15360GB: 6FC7D25B0h
23:16	NUSE	Namespace Utilization	support
24	NSFEAT	Namespace Features • Bit 4: 1b – OPTPERF: indicates that the fields NPWG, NPWA, NPDG, NPDA, and NOWS are defined for this namespace and should be used by the host for I/O optimization (refer to section 8.25); and NOWS defined for this namespace shall adhere to Optimal Write Size field setting defined in NVM Sets Attributes Entry (refer to Figure 251) for the NVM Set with which this namespace is associated. • Bit 1: 1b - NAWUN, NAWUPF, and NACWU should be used instead of the AWUN, AWUPF, and ACWU • Bit 0: 0h – not support this provisioning	12h
25	NLBAF	Number of LBA Formats (zero based value)	4h
26	FLBAS	Formatted LBA size • Bit 4: 0b - all of the metadata is transferred as a separate contiguous buffer of data. Only applicable when hosts using metadata. • Bits 3:0 - indicates LBA format index	depending on namespaces (Default=0h)
27	MC	Metadata Capabilities • Bit 1: 1b – support metadata as separate buffer	3h
28	DPC	End-to-end Data Protection Capabilities • Bit 4: 1b - support protection information transferred as the last eight bytes of metadata • Bit 2: 1b - support Protection Information Type 3 • Bit 1: 0b - not support Protection Information Type 2 • Bit 0: 0b - not support Protection Information Type 1	17h
29	DPS	End-to-end Data Protection Type Settings • 0b - Protection information is not enabled	depending on namespaces (Default=0h)
30	NMIC	Namespace Multi-path I/O and Namespace Sharing Capabilities • Bit 0: 0b - a private namespace and is able to be attached to only one controller at a time.	0h
31	RESCAP	Reservation Capabilities • 0b – not support the Reservation	0h
32	FPI	Format Progress Indicator • Bit 7: 1b - supports the Format Progress Indicator defined by bits 6:0 • Bit 6:0 - the percentage of the Format NVM, 0h means not in progress	80h
33	DLFEAT	• Bit 4 : 0b - indicates that the Guard field for the deallocated logical blocks that contain protection information is set to FFFFh • Bits 3 : 1b - indicates that the controller supports the Deallocate bit in the Write Zeroes command for this namespace	9h

		Bits 2:0 – 001b - A deallocated logical block returns all bytes cleared to 0h	
35:34	NAWU N	Namespace Atomic Write Unit Normal (zero based value) 1FFh – 512 LBA (2MB)	depending on namespaces (Default=1FFh)
37:36	NAWU PF	Namespace Atomic Write Unit Power Fail (zero based value) 0h – 1 LBA (4KB)	0h
39:38	NACW U	Namespace Atomic Compare & Write Unit (zero based value) 0h – not support	0h
41:40	NABSN	Namespace Atomic Boundary Size Normal	0h
43:42	NABO	Namespace Atomic Boundary Offset	0h
45:44	NABSP F	Namespace Atomic Boundary Size Power Fail	depending on namespaces (Default=0h)
47:46	NOIOB	Namespace Optimal I/O Boundary 0h - undefined	0h
63:48	NVMC AP	NVM Capacity	1920GB: 1920383410176 3840GB: 3840755982336 7680GB: 7681501126656 15360GB: 15363002253316
65:64	NPWG	Namespace Preferred Write Granularity	depending on namespaces (Default=0h)
67:66	NPWA	Namespace Preferred Write Alignment	depending on namespaces (Default=0h)
69:68	NPDG	Namespace Preferred Deallocate Granularity	depending on namespaces (Default=0h)
71:70	NPDA	Namespace Preferred Deallocate Alignment	depending on namespaces (Default=0h)
73:72	NOWS	Namespace Optimal Write Size	depending on namespaces (Default=0h)
95:92	ANAGR PID	ANA Group Identifier	0h
99	NSATT R	Namespace Attributes	0h
101:1 00	NVMSE TID	NVM Set Identifier	0h
103:1 02	ENDGI D	Endurance Group Identifier	0h
119:1 04	NGUID	Namespace Globally Unique Identifier	Variables ex) 2AE0AA97F000007 1ACE42E0010000016
127:1 20	EUI64	IEEE Extended Unique Identifier	Variables ex) ACE42E0010000016
131:1 28	LBAF0	LBA Format 0 Support	512B
135:1 32	LBAF1	LBA Format 1 Support	512B + 8B
139:1 36	LBAF2	LBA Format 2 Support	4KiB
143:1 40	LBAF3	LBA Format 3 Support	4KiB + 8B

147:1 44	LBAF4	LBA Format 4 Support	4KiB + 64B
...	...	...	...

Table 4-3 Identify Fields List

4.2 NVM I/O Command Set

Opcode	Command Name
00h	Flush
01h	Write
02h	Read
04h	Write Uncorrectable
08h	Write Zeroes
09h	Dataset Management

Table 4-4 NVM I/O Command List

4.3 NVM-MI Command Set

Category	Features	O/M	Supported
I2C Access	Refer to NVMe MI Basic Mgmt Specification	M	Y
	VPD Read	M	Y
Admin Over Out-of-Band NVMe-MI (MCTP over SMBus, MCTP over PCI VDM)	Set/Get Features	M	Y
	Get Log Page	M	Y
	Identify	M	Y
	Device Self-test	O	N
	Firmware Activate/Commit	O	Y
	Firmware Image Download	O	Y
	Format NVM	O	Y
	Namespace attach/detach	O	Y
	Namespace management (Create/Delete)	O	Y
	Sanitize	O	Y
	Security Receive	O	Y
	Security Send	O	Y
MI Commands (MCTP over SMBus, MCTP over PCI VDM)	Get Configuration	M	Y
	Set Configuration	M	Y
	Controller Health Status Poll	M	Y
	NVMe Subsystem Health Status Poll	M	Y
	Read NVMe-MI Data Structure	M	Y
	VPD Read	M	Y
	VPD Write	M	Y
	Reset	O	N

Table 4-5 NVMe-MI Command Set

4.4 SMART Health Information

Bytes	Default	Description
0	0	Critical Warning • Bit 7:5 – Reserved • Bit 4 – 1h: the volatile memory backup device has failed. (only valid if the controller has a volatile memory backup solution) • Bit 3 – 1h: the media has been placed in read only mode • Bit 2 – 1h: the device reliability has been degraded due to significant media related errors or any internal error that degrades device

		reliability • Bit 1 – 1h: the temperature has exceeded a critical threshold • Bit 0 – 1h: the available spare space has fallen below the threshold
2:1	Current temp	Temperature
3	100	Available Spare
4	10	Available Spare Threshold
5	0	Percentage Used
6	0	Endurance Group Critical Warning Summary
31:7	-	Reserved
47:32	0	Data Units Read
63:48	0	Data Units Written
79:64	0	Host Read Commands
95:80	0	Host Write Commands
111:96	0	Controller Busy Time
127:112	0	Power Cycles
143:128	0	Power On Hours
159:144	0	Unsafe Shutdowns
175:160	0	Media Errors
191:176	0	Number of Error Information Log Entries
511:192	-	Reserved
195:192	0	Warning Composite Temp Time
199:196	0	Critical Composite Temp Time
201:200	Current temp	Temp Sensor 1
203:202	Current temp	Temp Sensor 2
205:204	Current temp	Temp Sensor 3
511:204	0	Reserved

Table 4-6 SMART Fields

4.5 Out of Band Management

4.5.1 Basic Management Commands

Code	Bytes	Description	Value
0	00	Length of Status Indicates number of additional bytes to read before encountering PEC. Always 6 (06h) in this version of the spec.	06h
	01	Status Flags (SFLGS) This field indicates the status of the NVM subsystem. SMBus Arbitration – Bit 7 is set ‘1’ after a SMBus block read is completed all the way to the stop bit without bus contention and cleared to ‘0’ if a SMBus Send Byte FFh is received on this SMBus slave address. Drive Not Ready – Bit 6 is set to ‘1’ when the subsystem cannot process NVMe management commands, and the rest of the transmission may be invalid. If cleared to ‘0’, then the NVM subsystem is fully powered and ready to respond to management commands. This logic level intentionally identifies and prioritizes powered up and ready drives over their powered off neighbors on the same SMBus segment. Drive Functional – Bit 5 is set to ‘1’ to indicate an NVM subsystem is functional. If cleared to ‘0’, then there is an	Variables ex) Bit 7: 1b Complete Block Read Bit 6: 0b Device is Ready Bit 5: 1b Device Functional Bit 4: 1b Reset not Required Bit 3: 1b Port0 Link Up

		unrecoverable failure in the NVM subsystem and the rest of the transmission may be invalid. Reset Not Required - Bit 4 is set to '1' to indicate the NVM subsystem does not need a reset to resume normal operation. If cleared to '0' then the NVM subsystem has experienced an error that prevents continued normal operation. A controller reset is required to resume normal operation. Port 0 PCIe Link Active - Bit 3 is set to '1' to indicate the first port's PCIe link is up (i.e., the Data Link Control and Management State Machine is in the DL_Active state). If cleared to '0', then the PCIe link is down. Port 1 PCIe Link Active - Bit 2 is set to '1' to indicate the second port's PCIe link is up. If cleared to '0', then the second port's PCIe link is down or not present. Bits 1-0 shall be set to '1'.	Bit 2: 0b Port1 Link Down (DP not supported) Bit 1: 1b Bit 0: 1b
	02	SMART Warnings This field shall contain the Critical Warning field (byte 0) of the NVMe SMART / Health Information log. Each bit in this field shall be inverted from the NVMe definition (i.e., the management interface shall indicate a '0' value while the corresponding bit is '1' in the log page). Refer to the NVMe specification for bit definitions. If there are multiple controllers in the NVM subsystem, the management endpoint shall combine the Critical Warning field from every controller such that a bit in this field is: Cleared to '0' if any controller in the subsystem indicates a critical warning for that corresponding bit. Set to '1' if all controllers in the NVM subsystem do not indicate a critical warning for the corresponding bit.	Variables Depend on Critical Warning in NVMe/SMART Health Log
	03	Composite Temperature (CTemp) This field indicates the current temperature in degrees Celsius. If a temperature value is reported, it should be the same temperature as the Composite Temperature from the SMART log of hottest controller in the NVM subsystem. The reported temperature range is vendor specific, and shall not exceed the range -60 to +127°C. The 8 bit format of the data is shown below. This field should not report a temperature when that is older than 5 seconds. If recent data is not available, the NVMe management endpoint should indicate a value of 80h for this field. Value: Description • 00h-7Eh: Temperature is measured in degrees Celsius (0 to 126C) • 7Fh: 127C or higher • 80h: No temperature data or temperature data is more the 5 seconds old. • 81h: Temperature sensor failure • 82h-C3h: Reserved • C4: Temperature is -60C or lower • C5-FFh: Temperature measured in degrees Celsius is represented in twos complement (-1 to -59C)	Variables Depend on CTemp in NVMe/SMART Health Log

04	Percentage Drive Life Used (PDLU) Contains a vendor specific estimate of the percentage of NVM subsystem NVM life used based on the actual usage and the manufacturer's prediction of NVM life. If an NVM subsystem has multiple controllers the highest value is returned. A value of 100 indicates that the estimated endurance of the NVM in the NVM subsystem has been consumed, but may not indicate an NVM subsystem failure. The value is allowed to exceed 100. Percentages greater than 254 shall be represented as 255. This value should be updated once per power-on hour and equal the Percentage Used value in the NVMe SMART Health Log Page.	Variables Depend on Percentage Used in NVMe/SMART Health Log						
05	Current Over Temperature Warning Threshold (Optional) This field indicates the composite temperature over temperature warning threshold in degrees Celsius. This is intended to initially match the temperature reported in the WCTEMP field in the NVMe Identify Controller data structure. If the Over Temperature threshold for Composite Temperature is modified with set features, then the most recent value should be reported. The data format should match the same single byte format as the CTemp field with a range from -60 C to 127 C. A value of 0h means that this field is not reported or that the threshold is set to 0 C.	0h						
06	Current Power (Optional) This field reports the current NVM Subsystem power consumption. If both bit mapped fields are cleared to 0h, then this field is not reported. <table><tr><th>Bit</th><th>Definition</th></tr><tr><td>7</td><td>NVM Subsystem Idle (NVMSI): This bit is set to '1' when the NVM Subsystem is idle and has been idle for at least 5 s. Refer to the NVMe Idle Power (IDL P) definition.</td></tr><tr><td>6:0</td><td>NVM Subsystem Power (NVMSP): This field reports the ceiling function of the power consumed by the NVM Subsystem in watts. If the power consumed by the NVM Subsystem in watts is greater than or equal to 127 W, then 127 W is reported in this field. Power reported by the NVM Subsystem is determined in the following manner. If the NVMSI bit is set to '1', then the value returned in this field is: • equal to the value reported in the Idle Power (IDL P) field in the Power State </td></tr></table>	Bit	Definition	7	NVM Subsystem Idle (NVMSI): This bit is set to '1' when the NVM Subsystem is idle and has been idle for at least 5 s. Refer to the NVMe Idle Power (IDL P) definition.	6:0	NVM Subsystem Power (NVMSP): This field reports the ceiling function of the power consumed by the NVM Subsystem in watts. If the power consumed by the NVM Subsystem in watts is greater than or equal to 127 W, then 127 W is reported in this field. Power reported by the NVM Subsystem is determined in the following manner. If the NVMSI bit is set to '1', then the value returned in this field is: • equal to the value reported in the Idle Power (IDL P) field in the Power State	0h
Bit	Definition							
7	NVM Subsystem Idle (NVMSI): This bit is set to '1' when the NVM Subsystem is idle and has been idle for at least 5 s. Refer to the NVMe Idle Power (IDL P) definition.							
6:0	NVM Subsystem Power (NVMSP): This field reports the ceiling function of the power consumed by the NVM Subsystem in watts. If the power consumed by the NVM Subsystem in watts is greater than or equal to 127 W, then 127 W is reported in this field. Power reported by the NVM Subsystem is determined in the following manner. If the NVMSI bit is set to '1', then the value returned in this field is: • equal to the value reported in the Idle Power (IDL P) field in the Power State							

		Descriptor Data Structure for the corresponding NVMe power state if APW is set to a non-zero value; or • equal to the value reported in the Maximum Power (MP) field in the Power State Descriptor data structure for the corresponding NVMe power state, if the IDLP field is cleared to 0h. If the NVMSI bit is cleared to '0', then the value returned in this field is: • equal to the value reported in the Active Power Workload (APW) field in the Power State Descriptor Structure for the corresponding NVMe power state if the APW field is set to a non-zero value; or • equal to the value reported in the Maximum Power (MP) field in the Power State Descriptor data structure for the corresponding NVMe power state, if the APW field is cleared to 0h.	
	07	PEC An 8 bit CRC calculated over the slave address, command code, second slave address and returned data. Algorithm is in SMBus Specifications.	Variables ex) PEC = CAh
8	08	Length of identification Indicates number of additional bytes to read before encountering PEC. Always 22 (16h) in this version of the spec.	16h
	10:09	Vendor ID The 2 byte vendor ID, assigned by the PCI SIG. Should match VID in the Identify Controller command response. MSB is transmitted first.	1DC5h
	30:11	Serial Number 20 characters that match the serial number in the NVMe Identify Controller command response. First character is transmitted first	Variables
	31	PEC An 8 bit CRC calculated over the slave address, command code, second slave address and returned data. Algorithm is in SMBus Specifications	Variables ex) PEC = 35h
32	32	Length of GUID This is the number of bytes until the PEC code.	10h
	48:33	GUID This is a 16-byte Global Unique Identifier. MSB is transmitted first.	738920E56BEE4258 9A7ACEBDB35F0085h
	49	PEC An 8-bit CRC calculated over the slice address, command code,	Variables

		second slave address and returned data. Algorithm is defined in SMBus Specifications.	
50	50	Length of Telemetry Indicated the number of additional bytes to read before encountering PEC.	26h
	51	Temperature Flags This field reports the effect of temperature on the device's performance. Bit 7 - Temperature Throttling Set to 1b when the drive is throttling performance to prevent overheating. Clear to 0b when the device is not throttling. Bits 6-0 shall be set to 111111b.	Variables
	52	Max Power Supported This shall denote the max average power supported by this device rounded to the nearest watt. Some examples of how to use this is a 50W device is 0x32, a 25W device is 0x19, a 15W device is 0xF, an 8.25W device is 8W which is 0x8.	17h
	84:53	Configured Power State This is the power level entry that is currently set based on Set Features.	Variables
	88:85	Capacity This is the device raw capacity in GB in Hex (2048 GB in raw capacity = 0x800). Does not include any extra spare blocks within the NAND. MSB is transmitted first.	IDEMA Compliant Values
	89	PEC An 8-bit CRC calculated over the slice address, command code, second slave address and returned data. Algorithm in SMBus Specifications.	Variables
90	90	Length of Firmware Update Indicates number of additional bytes to read before encountering PEC.	04h
	91	Firmware Update Flags This field allows the host to control whether the current firmware allows new firmware images to be activated. Bit 7: Enable Firmware Update(Writable) * 1: Unlock Firmware Update - Drive shall enable Firmware update * 0: Lock Firmware Update - Drive shall block and error on Firmware download and activate commands Bit 6: Firmware Update Enabled(Read Only) * 1 Firmware Update Unlocked - Drive shall allow Firmware download and activate commands * 0 Firmware Update Locked - Drive shall block and error on Firmware download and activate commands Bits 5-0 shall be set to '1'.	FFh
	94:92	Reserved Shall be set to 00h	00h
	95	PEC An 8-bit CRC calculated over the slice address, command code, second slave address and returned data. Algorithm in SMBus Specifications.	Variables

96	96	Length of Firmware Version and Model Indicates number of additional bytes to read before encountering PEC.	38h
	104:97	Firmware Version Number This field shall indicate the activated firmware version that is running on the device after the firmware activation took place. The format of this field shall be as defined in field Firmware Revision (FR) section 5.15.2.2 Identify Controller Data Structure of the NVMe specification version 1.4b.	Variables
	112:105	Security Version Number This is the Security Version Number of the currently running firmware image. The supplier increments this number any time a firmware includes a fix for a security issue or critical firmware fix that customer agrees rollback prevention is required. This is a copy of SMART-22 - Security Version Number.	Variables
	152:113	Model Number The reason for 40 bytes is to keep this consistent with NVMe that already has this field in the identify command.	Variables
	153	PEC An 8-bit CRC calculated over the slice address, command code, second slave address and returned data. Algorithm in SMBus Specifications.	Variables
154	154	Payload length of Command Code 154 Indicates number of additional bytes to read before encountering PEC. Shall be set to 0Ah.	0Ah
	155	Panic Rest Action See EREC-2 (Panic Reset Action).	00h
	156	Device Recovery Action 1 See EREC-3 (Device Recovery Action 1).	Variables
	164:157	Panic ID See EREC-4 (Panic ID).	Variables
	165	Device Recovery Action 2 See EREC-14 (Device Recovery Action 2).	00h
	166	PEC An 8-bit CRC calculated over the SMBus address, command code, second SMBus address and returned data. The algorithm is defined in the SMBus Specification.	Variables
167	167	Payload length of Command Code 166 Indicates number of additional bytes to read before encountering PEC. Shall be set to 20h.	20h
	199:168	Panic Context Buffer Vendor may record additional information about the Panic ID reported in Command Code 154. Any unused bytes shall be cleared to zero.	Variables
	200	PEC An 8-bit CRC calculated over the SMBus address, command code, second SMBus address and returned data. The algorithm is defined in the SMBus Specification.	Variables
201	241:201	Reserved Shall be set to 00h	00h
242	242	Length of Secure Boot Failure Reporting Feature Indicates number of additional bytes to read before encountering PEC.	04h
	243	Secure Boot Failure Reporting Feature Bit 7: Secure Boot Failure Reporting Supported When set to 0x1 the secure boot failure reporting is supported.	80h

		When set to 0x0 the secure boot failure reporting is not supported. Bit 6: Secure Boot Failure Status When set to 0b there is no secure boot failure. When set to 1b there is a secure boot failure. This bit shall only be set if the Secure Boot Failure Reporting Supported bit is set to 1b and there is a secure boot failure. Bit 5:0 Reserved. Shall be set to 0x0.	
	244	Recovery Code OCP Recovery/Platform Root-of-Trust for Recovery code.	0h
	246:245	Reserved Shall be set to 00h	00h
	247	PEC An 8-bit CRC calculated over the slice address, command code, second slave address and returned data. Algorithm in SMBus Specifications.	Variables
248	248	Length of Data Format Version Indicates number of additional bytes to read before encountering PEC.	06h
	250:249	Data Format Version Number Indicates the version of this mapping used in the device. MSB is transmitted first.	0004h
	254:251	Reserved Shall be set to 00h	00h
	255	PEC An 8-bit CRC calculated over the slice address, command code, second slave address and returned data. Algorithm in SMBus Specifications.	Variables

Table 4-7 Basic Management Command

4.5.2 VPD

Byte	Name
07:00	Common Header
119:08	Product Info Area (optional)
135:120	Board Info Area (optional)
Vendor Specific:136	MultiRecord Info Area
Vendor Specific	Internal Use Area (optional)
Vendor Specific	Chassis Info Area (optional)

Table 4-8 Vital Product Data Structure (VPD Elements)

Bytes	Description	Value
00	IPMI Format Version Number (IPMIVER) This field indicates the IPMI Format Version.	01h
01	Internal Use Area Starting Offset (UAOFF) This field indicates the starting offset in multiples of 8 bytes for the Internal Use Area. A value of 0h may be used to indicate the Internal Use Area is not present.	00h
02	Chassis Info Area Starting Offset (CIAOFF) This field indicates the starting offset in multiples of 8 bytes for the	00h

	Chassis Info Area. A value of 0h may be used to indicate the Chassis Info Area is not present.	
03	Board Info Area Starting Offset (BIAOFF) This field indicates the starting offset in multiples of 8 bytes for the Board Info Area. A value of 0h may be used to indicate the Board Info Area is not present.	0Fh
04	Product Info Area Starting Offset (PIAOFF) This field indicates the starting offset in multiples of 8 bytes for the Product Info Area.	01h
05	MultiRecord Info Area Starting Offset (MRIOFF) This field indicates the starting offset in multiples of 8 bytes for the MultiRecord Info Area.	11h
06	Reserved	00h
07	Common Header Checksum (CHCHK) Checksum computed over byte 0 to byte 6. The checksum is computed by adding the 8-bit value of the bytes modulo 256 and then taking the 2's complement of this sum. When the checksum and the sum of the bytes module 256 are added, the result should be 0h.	DEh

Table 4-9 Common Header

Bytes	Description	Value
00	IPMI Format Version Number (IPMIVER) This field indicates the IPMI Format Version.	01h
01	Product Info Area Length (PALEN) This field indicates the length of the Product Info Area in multiples of 8 bytes.	0Eh
02	Language Code (LCODE) This field indicates the language used. A value of 19h is used to indicate English.	19h
03	Manufacturer Name Type/Length (MNTL) This field indicates the type and length of the Manufacturer Name field. The maximum length is 8.	C8h
11:04	Manufacturer Name (MNAME) This field indicates the Manufacturer name in 8-bit ASCII. Unused bytes should be NULL characters. This Manufacturer name in this field should correspond to that in the PCI Subsystem Vendor ID (SSVID) and IEEE OUI Identifier fields in the Identify Controller Data Structure.	
12	Product Name Type/Length (PNTL)	D8h

	This field indicates the type and length of the Product Name field. The maximum length is 24.	
36:13	Product Name (PNAME) This field indicates the Product name in 8-bit ASCII. Unused bytes should be NULL characters.	
37	Product Part/Model Number Type/Length (PPMNNTL) This field indicates the type and length of the Product Part/Model Number field. The maximum length is 40.	E8h
77:38	Product Part/Model Number (PPMN) This field indicates the Product Part/Model Number in 8-bit ASCII. Unused bytes should be NULL characters. This field should contain the same value as the Model Number (NM) field in the NVMe Identify Controller Data Structure.	Variables
78	Product Version Type/Length (PVTL) This field indicates the type and length of the Product Part/Model Number field. The maximum length is 2.	C2h
80:79	Product Version (PVER) This field indicates the Product Version in 8-bit ASCII. Unused bytes should be NULL characters.	30h 00h
81	Product Serial Number Type/Length (PSNTL) This field indicates the type and length of the Product Serial Number field. The maximum length is 20.	D4h
101:82	Product Serial Number (PSN) This field indicates the Product Serial Number in 8-bit ASCII. Unused bytes should be NULL characters. This field should contain the same value as the Serial Number (SN) field in the NVMe Identify Controller Data Structure.	Variables
102	Asset Tag Type/Length (ATTL) This field indicates the type and length of the Asset Tag field. A value of 0h may be used to indicate an Asset Tag is not present.	00h
103	FRU File ID Type/Length (FFTL) This field indicates the type and length of the FRU File ID field. A value of 0h may be used to indicate a FRU File ID is not present.	00h
104	End of Record (EOR) A value of C1h in this field indicates the end of record.	C1h
110:105	Reserved	00h
111	Product Info Area (PICK)	Variables

Table 4-10 Product Info Area (Offset 8 Bytes)

Bytes	Description	Value
00	IPMI Format Version Number (IPMIVER) This field indicates the IPMI Format Version.	01h
01	Board Info Area Length (PALEN) This field indicates the length of the Board Info Area in multiples of 8 bytes.	02h
02	Language Code (LCODE) This field indicates the language used. A value of 19h is used to indicate English.	19h
05:03	Manufacturing Date/Time	00h
06	Manufacturer Name Type/Length (MNLT) This field indicates the type and length of the Manufacturer Name field. The maximum length is 8.	C0h
07	Product Name Type/Length (PNTL) This field indicates the type and length of the Product Name field. The maximum length is 24.	C0h
08	Product Serial Number Type/Length (PSNTL) This field indicates the type and length of the Product Serial Number field. The maximum length is 20.	C0h
09	Product Part/Model Number Type/Length (PPMNLT) This field indicates the type and length of the Product Part/Model Number field. The maximum length is 40.	C0h
10	FRU File ID Type/Length (FFTL) This field indicates the type and length of the FRU File ID field. A value of 0h may be used to indicate a FRU File ID is not present.	C0h
11	End of Record (EOR) A value of C1h in this field indicates the end of record.	C1h
14:12	Reserved	00h
15	Board Info Area (BCHK)	63h

Table 4-11 Board Info Area (Offset 120 Bytes)

Bytes	Description	Value
00	NVMe Record Type ID	0Bh
01	Record Format	02h
	Bit	
	Definition	
	7	Set to '1' if last record in list
	6:0	Record format version = 2.
02	Record Length (RLEN)	3Bh

	This field indicates the length of the MultiRecord Area in bytes without including the first 5 bytes that are common to all MultiRecords.	
03	Record Checksum This field is used to give the record data a zero checksum (i.e., the modulo 256 sum of the record data bytes from byte offset 05 to the end of this record plus this checksum byte equals zero).	Variables
04	Header Checksum This field is used to give the record header a zero checksum (i.e., the modulo 256 sum of the first byte of the header through this checksum byte equals zero).	Variables
05	NVMe MultiRecord Area Version Number This field indicates the version number of this NVMe MultiRecord. This field shall be cleared to 0h in this version of the specification.	00h
06	Form Factor This field indicates the form factor of the Management Endpoint.	11h
12:07	Reserved	00h
13	Initial 1.8V Power Supply Requirements This field specifies the initial 1.8 V power supply requirements in Watts prior to receiving a Set Slot Power message.	00h
14	Maximum 1.8V Power Supply Requirements This field specifies the maximum 1.8 V power supply requirements in Watts.	00h
15	Initial 3.3V Power Supply Requirements This field specifies the initial 3.3 V power supply requirements in Watts prior to receiving a Set Slot Power message.	00h
16	Maximum 3.3V Power Supply Requirements This field specifies the maximum 3.3 V power supply requirements in Watts.	00h
17	Reserved	00h
18	Maximum 3.3V aux Power Supply Requirements This field specifies the maximum 3.3 V power supply requirements in 10 Mw units.	03h
19	Initial 5V Power Supply Requirements This field specifies the initial 5 V power supply requirements in Watts prior to receiving a Set Slot Power message.	00h
20	Maximum 5V Power Supply Requirements This field specifies the maximum 5 V power supply requirements in Watts.	00h
21	Initial 12V Power Supply Requirements This field specifies the initial 12 V power supply requirements in Watts	17h

	prior to receiving a Set Slot Power message.	
22	Maximum 12V Power Supply Requirements This field specifies the maximum 12 V power supply requirements in Watts.	17h
23	Maximum Thermal Load This field specifies the maximum thermal load from the NVM Subsystem in Watts.	17h
36:24	Total NVM Capacity This field indicates the total NVM capacity of the NVM Subsystem in bytes. If the NVM Subsystem supports Namespace Management, then this field should correspond to the value reported in the TNVMCAP field in the NVMe Identify Controller Data Structure. A value of 0h may be used to indicate this feature is not supported.	Variables
63:37	Reserved	00h

Table 4-12 NVMe MultiRecord Area (Offset 136 Bytes)

Bytes	Description		Value
00	NVMe PCIe Port Record Type ID		0Ch
01	Record Format		02h
	Bit	Definition	
	7	Set to '1' if last record in list	
	6:0	Record format version = 2.	
02	Record Length (RLEN) This field indicates the length of the MultiRecord Area in bytes without including the first 5 bytes that are common to all MultiRecords.		0Bh
03	Record Checksum This field is used to give the record data a zero checksum (i.e., the modulo 256 sum of the record data bytes from byte offset 05 to the end of this record plus this checksum byte equals zero).		Variables
04	Header Checksum This field is used to give the record header a zero checksum (i.e., the modulo 256 sum of the first byte of the header through this checksum byte equals zero).		Variables
05	NVMe PCIe Port MultiRecord Area Version Number This field indicates the version number of this NVMe PCIe Port MultiRecord. This field shall be set to 1h in this version of the specification.		01h

06	PCIe Port Number This field contains the PCIe port number. This is the same value as that reported in the Port Number field in the PCIe Link Capabilities Register.	00h														
07	Port Information This field indicates information about the PCIe Ports in the device. Bits 7:1 are reserved. Bit 0, if set to '1' indicates that all PCIe ports within the device have the same capabilities (i.e., the capabilities listed in this structure are consistent across each PCIe port).	01h														
08	PCIe Link Speed This field indicates a bit vector of link speeds supported by the PCIe port. <table><tr><th>Bit</th><th>Definition</th></tr><tr><td>7:5</td><td>Reserved</td></tr><tr><td>4</td><td>Set to '1' if the PCIe link supports 32.0 GT/s, otherwise cleared to '0'.</td></tr><tr><td>3</td><td>Set to '1' if the PCIe link supports 16.0 GT/s, otherwise cleared to '0'.</td></tr><tr><td>2</td><td>Set to '1' if the PCIe link supports 8.0 GT/s, otherwise cleared to '0'.</td></tr><tr><td>1</td><td>Set to '1' if the PCIe link supports 5.0 GT/s, otherwise cleared to '0'.</td></tr><tr><td>0</td><td>Set to '1' if the PCIe link supports 2.5 GT/s, otherwise cleared to '0'.</td></tr></table>	Bit	Definition	7:5	Reserved	4	Set to '1' if the PCIe link supports 32.0 GT/s, otherwise cleared to '0'.	3	Set to '1' if the PCIe link supports 16.0 GT/s, otherwise cleared to '0'.	2	Set to '1' if the PCIe link supports 8.0 GT/s, otherwise cleared to '0'.	1	Set to '1' if the PCIe link supports 5.0 GT/s, otherwise cleared to '0'.	0	Set to '1' if the PCIe link supports 2.5 GT/s, otherwise cleared to '0'.	1Fh
Bit	Definition															
7:5	Reserved															
4	Set to '1' if the PCIe link supports 32.0 GT/s, otherwise cleared to '0'.															
3	Set to '1' if the PCIe link supports 16.0 GT/s, otherwise cleared to '0'.															
2	Set to '1' if the PCIe link supports 8.0 GT/s, otherwise cleared to '0'.															
1	Set to '1' if the PCIe link supports 5.0 GT/s, otherwise cleared to '0'.															
0	Set to '1' if the PCIe link supports 2.5 GT/s, otherwise cleared to '0'.															
09	PCIe Maximum Link Width The maximum PCIe link width for this NVM Subsystem port. This is the expected negotiated link width that the port link trains to if the platform supports it. A Requester may compare this value with the PCIe Negotiated Link Width to determine if there has been a PCIe link training issue. <table><tr><th>Value</th><th>Definition</th></tr><tr><td>0</td><td>Reserved</td></tr><tr><td>1</td><td>PCIe x1</td></tr><tr><td>2</td><td>PCIe x2</td></tr></table>	Value	Definition	0	Reserved	1	PCIe x1	2	PCIe x2	04h						
Value	Definition															
0	Reserved															
1	PCIe x1															
2	PCIe x2															

	3	Reserved		
	4	PCIe x4		
	5 to 7	Reserved		
	8	PCIe x8		
	9 to 11	Reserved		
	12	PCIe x12		
	13 to 15	Reserved		
	16	PCIe x16		
	17 to 31	Reserved		
	32	PCIe x32		
	33 to 255	Reserved		
10	MCTP Support This field contains a bit vector that specifies the level of support for the NVMe Management Interface. Bits 7:1 are reserved. Bit 0, if set to ‘1’ indicates that MCTP-based management commands are supported on the PCIe port.			01h
11	Ref Clk Capability This field contains a bit vector that specifies the PCIe clocking modes supported by the port.			01h
12	Port Identifier This field contains the NVMe-MI Port Identifier.			00h
15:13	Reserved			00h

Table 4-13 NVMe PCle Port MultiRecord Area (Offset 200 Bytes)

Element	Bytes	Description	Value	
Topology MultiRecord Area Header	00	Topology Record Type ID	0Dh	
	01	Record Format:		82h
		Bit	Definition	
		7	Set to '1' if last record in list.	
		6:0	Record format version = 2.	
	02	Record Length (RLEN): This field indicates the length of the MultiRecord Area in bytes without including the first 5 bytes that are common to all MultiRecords.		15h
03	Record Checksum: This field is used to give the record data a zero checksum (i.e., the modulo 256 sum of the record data bytes from byte offset 05 to the end of this record plus this checksum byte equals zero).		Variables	

	04	Header Checksum: This field is used to give the record header a zero checksum (i.e., the modulo 256 sum of the first byte of the header through this checksum byte equals zero).	Variables						
	05	Version Number: This field indicates the version number of this Topology MultiRecord. This field shall be cleared to 0h in this version of the specification.	00h						
	06	Reserved	00h						
	07	Element Count (N): This field indicates the number of Element Descriptors in this Topology MultiRecord. The value of 0h is reserved.	02h						
NVM Subsystem Element Descriptor	08	Type: This field indicates the type of the Element Descriptor.	07h						
	09	Revision: This field indicates the revision of the Element Descriptor.	00h						
	10	Length: This field indicates the length of the NVM Subsystem Element Descriptor in bytes.	0Ch						
	11	SMBus/I2C Address Info: If the NVM Subsystem supports an MCTP over SMBus/I2C port, then this field indicates the SMBus/I2C address for MCTP over SMBus/I2C port and whether or not SMBus ARP is supported; otherwise this field has a value of 0h. <table><tr><th>Bit</th><th>Definition</th></tr><tr><td>7:1</td><td>SMBus/I2C Address: This field contains the 7-bit SMBus/I2C address.</td></tr><tr><td>0</td><td>ARP Capable: This bit is set to '1' if SMBus ARP is supported, else it is cleared to '0'.</td></tr></table>	Bit	Definition	7:1	SMBus/I2C Address: This field contains the 7-bit SMBus/I2C address.	0	ARP Capable: This bit is set to '1' if SMBus ARP is supported, else it is cleared to '0'.	3Bh
	Bit	Definition							
7:1	SMBus/I2C Address: This field contains the 7-bit SMBus/I2C address.								
0	ARP Capable: This bit is set to '1' if SMBus ARP is supported, else it is cleared to '0'.								
12	SMBus/I2C Capabilities: If the NVM Subsystem supports an SMBus/I2C port then this field indicates the SMBus/I2C capabilities; otherwise this field has a value of 0h. <table><tr><th>Bit</th><th>Description</th></tr><tr><td>7</td><td>Reset: This bit is set to '1' if all of the SMBus/I2C reset mechanisms are supported as defined by the associated form factor specification. This bit is cleared to '0' if the form factor does not define SMBus Reset or the NVMe Storage Device does not support all of the SMBus/I2C reset mechanisms defined by the specification for the Form Factor in the Host Connector Element Descriptor.</td></tr></table>	Bit	Description	7	Reset: This bit is set to '1' if all of the SMBus/I2C reset mechanisms are supported as defined by the associated form factor specification. This bit is cleared to '0' if the form factor does not define SMBus Reset or the NVMe Storage Device does not support all of the SMBus/I2C reset mechanisms defined by the specification for the Form Factor in the Host Connector Element Descriptor.	82h			
Bit	Description								
7	Reset: This bit is set to '1' if all of the SMBus/I2C reset mechanisms are supported as defined by the associated form factor specification. This bit is cleared to '0' if the form factor does not define SMBus Reset or the NVMe Storage Device does not support all of the SMBus/I2C reset mechanisms defined by the specification for the Form Factor in the Host Connector Element Descriptor.								

		6:2	Reserved		
		1:0	Maximum Speed: This field is set to the highest supported SMBus/I2C clock speed.		
			Value	Description	
			0	100 kHz	
			1	400 kHz	
			2	1 MHz	
			3	Reserved	
	13	NVM Subsystem Port Descriptor Count: This field indicates the number of NVM Subsystem Port Descriptors associated with the NVM Subsystem. The permitted range of values is 1 to 64.			01h
	14	Type: This field indicates the type of an NVM Subsystem Port Descriptor. The NVM Subsystem Port Descriptor Type is 0.			00h
	15	Length: This field indicates the length of the NVM Subsystem Port Descriptor in bytes.			06h
	16	PCIe Link Speed: This field indicates a bit vector of link speeds supported by the PCIe port.			1Fh
		Bit	Description		
		7:5	Reserved		
		4	Set to '1' if the PCIe link supports 32.0 GT/s, otherwise cleared to '0'.		
		3	Set to '1' if the PCIe link supports 16.0 GT/s, otherwise cleared to '0'.		
		2	Set to '1' if the PCIe link supports 8.0 GT/s, otherwise cleared to '0'.		
1		Set to '1' if the PCIe link supports 5.0 GT/s, otherwise cleared to '0'.			
17	PCIe Maximum Link Width: The maximum PCIe link width for this NVM Subsystem port.			04h	
	Value	Description			
	0	Reserved			
	1	PCIe x1			
	2	PCIe x2			
	3	Reserved			

		4	PCIe x2				
		5 to 7	Reserved				
		8	PCIe x2				
		9 to 11	Reserved				
		12	PCIe x12				
		13 to 15	Reserved				
		16	PCIe x16				
		17 to 31	Reserved				
		32	PCIe x32				
		33 to 255	Reserved				
	18	RefClk Capability: This field contains a bit vector that specifies the PCIe clocking modes supported by the port.					01h
		Bit	Description				
		7:4	Reserved				
		3	Set to '1' if the device automatically uses RefClk if provided and otherwise uses SRIS, otherwise cleared to '0'.				
		2	Set to '1' if the PCIe link supports Separate RefClk with SSC (SRIS), otherwise cleared to '0'.				
		1	Set to '1' if the PCIe link supports Separate RefClk with no SSC (SRNS), otherwise cleared to '0'.				
		0	Set to '1' if the PCIe link supports common RefClk, otherwise cleared to '0'.				
19	Port Identifier: This field contains the NVMe-MI Port Identifier associated with this port.				00h		
FRU Information Device Element	20	Type: This field indicates the type of the Element Descriptor.				08h	
	21	Revision: This field indicates the revision of the Element Descriptor.				00h	
	22	Length: This field indicates the length of the NVM Subsystem Element Descriptor in bytes.				06h	
	23	SMBus/I2C Address Info: If the NVM Subsystem supports an MCTP over SMBus/I2C port, then this field indicates the SMBus/I2C address for MCTP over SMBus/I2C port and whether or not SMBus ARP is supported; otherwise this field has a value of 0h.				A6h	
Bit		Definition					

		7:1	SMBus/I2C Address: This field contains the 7-bit SMBus/I2C address.			
		0	ARP Capable: This bit is set to '1' if SMBus ARP is supported, else it is cleared to '0'.			
	24	SMBus/I2C Capabilities: If the NVM Subsystem supports an SMBus/I2C port then this field indicates the SMBus/I2C capabilities; otherwise this field has a value of 0h.				02h
		Bit	Description			
		7	Reset: This bit is set to '1' if all of the SMBus/I2C reset mechanisms are supported as defined by the associated form factor specification. This bit is cleared to '0' if the form factor does not define SMBus Reset or the NVMe Storage Device does not support all of the SMBus/I2C reset mechanisms defined by the specification for the Form Factor in the Host Connector Element Descriptor.			
		6:2	Reserved			
		1:0	Maximum Speed: This field is set to the highest supported SMBus/I2C clock speed.			
			Value	Description		
			0	100 kHz		
			1	400 kHz		
2	1 MHz					
3	Reserved					
25	Maximum FRU Information Device Size				08h	

Table 4-14 Topology MultiRecord Area (Offset 216 Bytes)

5.0 Security

5.1 Feature Overview

Category	Features	Note
Secure Boot / Update	RSA4096-based FW image authentication Supported	
	Anti-Rollback Protection Supported	
	SHA2-512 Hash algorithm Supported	
	Up-to 31 Security version update Supported	
	Up-to 5 Keys revocation Supported	
User Data Encryption	AES XTS 256 bits HW crypto engine	IP supplier: Synopsys
NVMe	Security Receive command Supported	
	Security Send command Supported	
TCG OPAL	OPAL SSC v2.01 Supported	
	Command Buffer Size: 8KB	
TCG OPAL Feature Set	Additional Datastore v.1.0	
	Single User Mode v1.0	
	Block SID Authentication v1.0	
	PSID v1.0	
Crypto Erase	Via both (1) NVMe Format (Secure Erase Setting) and (2) TCG GenKey/Revert method Supported	TRNG used for key generation
		TRNG IP supplier: ARM
Compliance		

5.2 TCG Level-0 Discovery Table

5.2.1 TPer Feature (Feature Code = 0x0001)

Bytes	Bit	Field	Value
01:00	15:0	Feature Code	0x0001
02	7:4	Version	0x1
03	7:0	Length	0x0C
04	6	ComID Mgmt Supported	0
	4	Streaming Supported	1
	3	Buffer Mgmt Supported	0
	2	ACK/NAK Supported	0
	1	Async Supported	0
	0	Sync Supported	1

5.2.2 Locking Feature (Feature Code = 0x0002)

Bytes	Bit	Field	Value
01:00	15:0	Feature Code	0x0002
02	7:4	Version	0x1
03	7:0	Length	0x0C
04	5	MBR Done	0
	4	MBR Enabled	0
	3	Media Encryption	1
	2	Locked	0
	1	Locking Enabled	0
	0	Locking Supported	1

5.2.3 Geometry Reporting Feature (Feature Code = 0x0003)

Bytes	Bit	Field	Value
01:00	15:0	Feature Code	0x0003
02	7:4	Version	0x1
03	7:0	Length	0x1C
15:12	31:0	LogicalBlockSize	0x1000
23:16	63:0	AlignmentGranularity	0x1
31:24	63:0	LowestAlignedLBA	0x00

5.2.4 Single User Mode Feature (Feature Code = 0x0201)

Bytes	Bit	Field	Value
01:00	15:0	Feature Code	0x0201
02	7:4	Version	0x1
03	7:0	Length	0x0C
07:04	31:0	Number of Locking Objects Supported	0x09
08	2	Policy	1
	1	All	0
	0	Any	0

5.2.5 DataStore Table Feature (Feature Code = 0x0202)

Bytes	Bit	Field	Value
01:00	15:0	Feature Code	0x0202
02	7:4	Version	0x1
03	7:0	Length	0x0C
07:06	15:0	Maximum number of DataStore tables	0x9
11:08	31:0	Maximum total size of DataStore tables	0xA00000
15:12	31:0	DataStore table size alignment	0x1

5.2.6 Opal SSC V2.00 Feature (Feature Code = 0x0203)

Bytes	Bit	Field	Value
01:00	15:0	Feature Code	0x0203
02	7:4	Version	0x1
03	7:0	Length	0x10

05:04	15:0	Base ComID	0x07FE
07:06	15:0	Number of ComIDs	1
08	0	Range Crossing Behavior	0x1
10:09	15:0	Number of Locking SP Admin Authorities	0x04
12:11	15:0	Number of Locking SP User Authorities	0x09
13	7:0	Initial C_PIN_SID PIN Indicator	0x00
14	7:0	Behavior of C_PIN_SID PIN upon TPer Revert	0x00

5.2.7 Block SID Authentication Feature (Feature Code = 0x0402)

Bytes	Bit	Field	Value
01:00	15:0	Feature Code	0x0402
02	7:4	Version	0x1
03	7:0	Length	0x0C
04	1	SID Blocked State	0
	0	SID Value State	0
05	0	Hardware Reset	0

5.3 Secure Boot Failure Indication Method

5.3.1 PCIe Class Code

Secure boot failure could be detected by reading out PCIe class code in PCIe configuration header, Base Class Code of which will be set to **FFh** and its Sub-Class Code will be set to **00h** when secure boot failure occurs.

5.3.2 SMBus (NVMe Basic Management Command)

Secure boot failure could be detected by reading out secure boot failure reporting feature supported at 243 bytes offset of NVMe basic management command data format where bit 6 will be set to **1b** if there is secure boot failure.

6.0 Package Specification

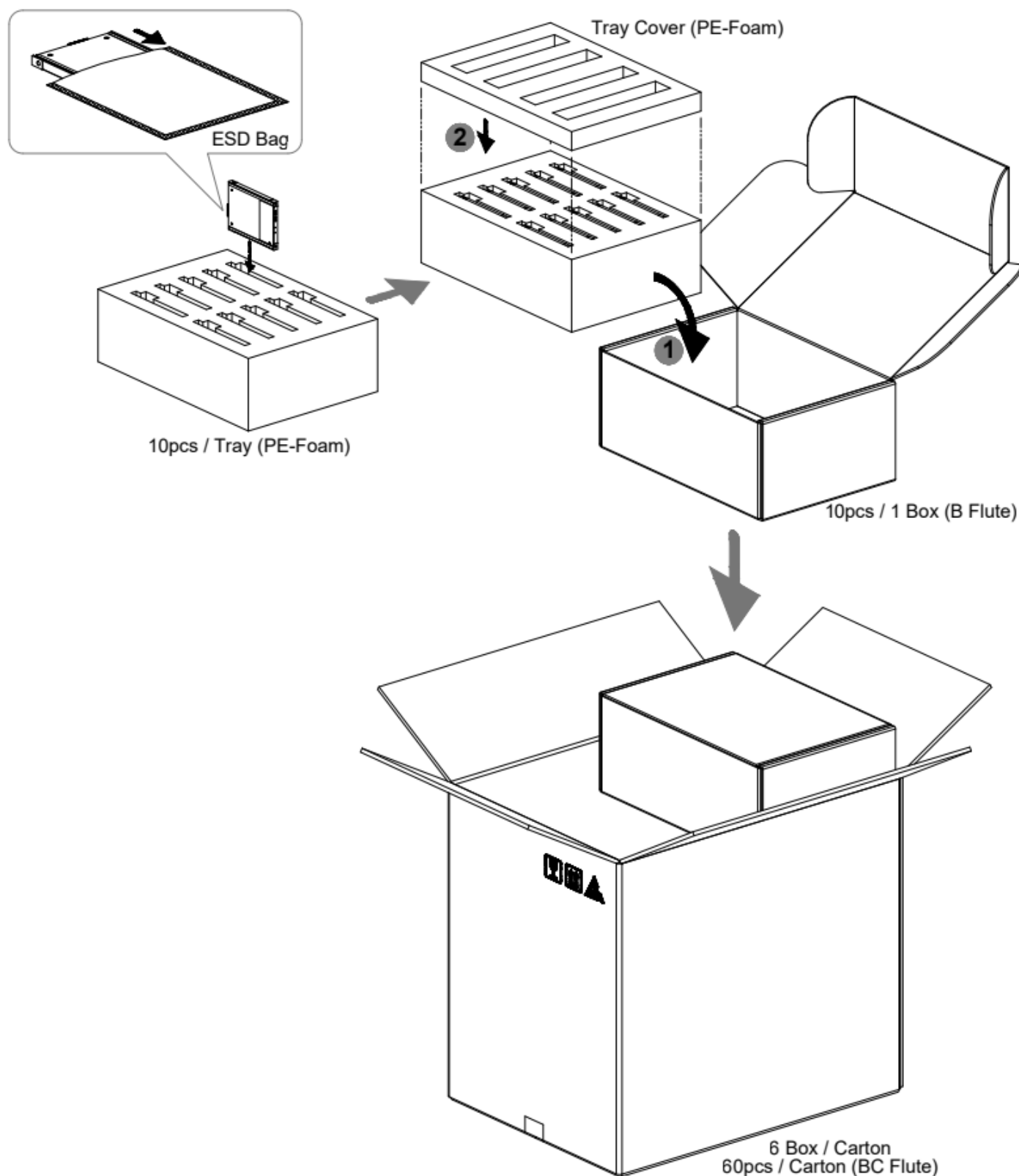


Figure 6-1 Package Specification

7.0 Ordering Information

Model Name	Capacity	DWPD	Type	Remark
TRUSTA T7P5-001T6T7E3S01D040	1600GB	3	E3.S PCIe Gen5	0°C~70°C
TRUSTA T7P5-001T9T5E3S01D040	1920GB	1	E3.S PCIe Gen5	
TRUSTA T7P5-003T2T7E3S01D040	3200GB	3	E3.S PCIe Gen5	
TRUSTA T7P5-003T8T5E3S01D040	3840GB	1	E3.S PCIe Gen5	
TRUSTA T7P5-006T4T7E3S01D040	6400GB	3	E3.S PCIe Gen5	
TRUSTA T7P5-007T6T5E3S01D040	7680GB	1	E3.S PCIe Gen5	
TRUSTA T7P5-012T8T7E3S01D040	12800GB	3	E3.S PCIe Gen5	
TRUSTA T7P5-015T3T5E3S01D040	15360GB	1	E3.S PCIe Gen5	

Table 7-1 Ordering Information